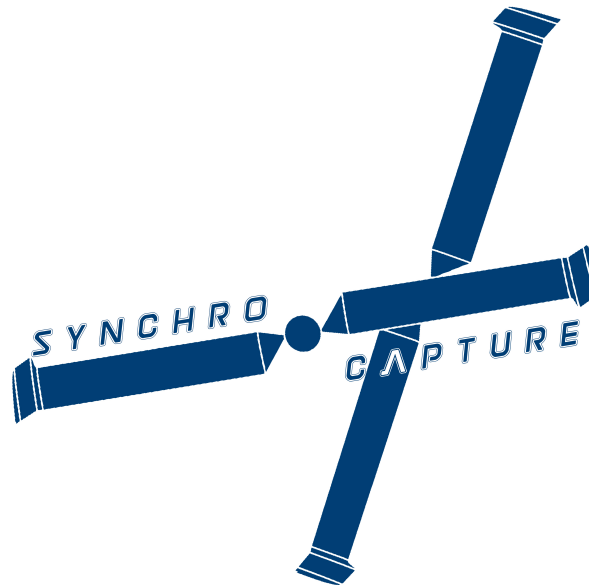


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Controller design of a Synchropter



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Abstract

As part of the year Group Design Project, design of 6Dof Helicopter was proposed to recover a wholly reusable launch system. This report starts with analysing the most suitable control methodology to be used for the helicopter controller. It presents the possible design and implementation of a Linear Quadratic Gaussian controller, assuming a linearized model of the Helicopter dynamics. More emphasis was given to the design of a sensor-based Kalman filter. The controller is not completed but the Kalman filter is able to cancel out noisy measurements. The observability of the system is also investigated.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
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Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
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Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
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†Sub-team leads are shown in **bold**

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Part I

Preliminary Design

1 Primary control objectives

The decision on the best control methodology is dependent on the aspect of control which is being prioritized: stability of the system or its response to control inputs. Generally, designing a controller for stability decreases the pilot's ability to control the vehicle and since the mission involves both an autonomous and remote controlled part, the decision was more complicated. On the one hand, less critical aspects of the mission such as take-off, climb, and cruise were set to be autonomous, meaning the helicopter's stability and its ability to follow a trajectory are essential. On the other hand, the most critical mission legs such as payload capture, payload cruise and landing were planned to be remote controlled, requiring a fully controllable/manoeuvrable helicopter at high descent rates (ideally > 3000 ft/min). The team was committed to address both aspects of the mission but the emphasis was given to stability since helicopters are inherently unstable.

2 Control methodology

A trade-off study on the most appropriate control methods for this case was carried-out following the *Survey on Flight Control Technology for Large-Scale Helicopter* [1], which analysed the relevance and limitations of classical, modern and intelligent control methodologies historically used on helicopter control design, as well as the continuous research on this field.

2.1 Classical control

PID-based (proportional–integral–derivative) feedback control is a relatively simple feedback type of control which considers the steady-state error between the system output and a reference value. This type of control was successfully tested on helicopters when modified and optimized (eg. scheduled Stability and Control Augmentation Systems) but proved to be insufficient on its own when it comes to highly coupled complex systems as helicopters.

Contrarily to PID, an Explicit Model Following Control approach enables input decoupling control but it is highly dependent on the model dynamics, particularly the modelling of the rotor. Given this, there is associated risk of robustness issues in case the model is not able to accurately represent the high frequency dynamics of the propellers.

Gain-scheduling control is adequate to non-linear systems and consists on the ability of adjusting its gains to different operating points. It is a major advantage when the mission requires different control responses and there isn't a single optimal performance strategy. However, all the references of application of this method mentioned on the survey, required a specific software package (CONDUIT- Control Designer's Unified Interface).

2.2 Intelligent control

Fuzzy control is a relatively new approach which uses artificial intelligence and fuzzy logic to copy the human thought processing. The survey concluded that the method is insufficient to stabilise an helicopter on its own. Moreover, its application is still being studied and so far there its performance as a controller are assessed purely based on virtual simulations. ANN (Artificial Neural Network)

control is based on Machine learning algorithms without necessarily relying on an exact model. Nonetheless, this approach is dependent on training strategies and large quantities of data.

2.3 Modern control

The article highlights two control methodologies for the fact that they have been successfully tested in piloted simulations: H_∞/H_2 control is a mathematically optimal control method that could be relevant to our case given its application to helicopter slung load control problems; INDI (Incremental Nonlinear Dynamic Inversion) control is sensor-based approach and an upgraded version of NDI. Important to note that both these methods are quite advanced.

Observer Feedback Control is a simple method to implement which enables the estimation of all states by means of an observer. The survey mentions the application of Kalman filters - for linear systems - and Extended Kalman filters - non-linear systems - to helicopter control problems. This strategy is supposed to enable precise state tracking. There might be robustness issues, which can be fixed with LTR (Loop-transfer recovery). The following control approaches were not explored as much on the article partly due to lack of relevant testing/application (eg. Backstepping and Sliding Mode control). In Eigenstructure Assignment Control, the eigenvalues and eigenvectors of the system are manipulated to obtain the desired response. Adaptive control enables the adaptation to time-varying parameters in a system. It is particularly interesting in trajectory tracking and the study stresses its increasing application to helicopters. Similarly to gain-scheduling and Adaptive control, LPV (Linear Parameter-varying) control is based on the design of one controller, to achieve similar results to Adaptive control and Gain-scheduling control. Similarly to LQG, Model Predictive Control also minimizes a cost function but anticipates future events through a different strategy. It is an advanced method and widely used, with increasing popularity in helicopter control. Table 2 presents the methodologies that were initially ruled out and summarizes the reasons why whereas table 3 displays the methods considered for the specific task and objectives.

Control Methodologies	Inadequate	Insufficient applied examples	High complexity/ difficult implementation
PID	X		
Gain-scheduling			X
Fuzzy			X
ANN			X
H_∞/H_2			X
INDI			X
Backstepping		X	
Sliding Mode		X	

Table 2: Excluded methodologies

Control Methodologies	Linear/ Non-linear	Level of complexity	Robustness
LQG	linear	low	not guaranteed
LPV	non-linear	high	-
Eigenstructure	linear	low	not guaranteed
Adaptive	both	high	yes
MPC	both	high	-

Table 3: Methodologies considered for the project and some relevant properties.

In the end the time constraint of the project was the most determinant aspect and led to the decision of LQG control. Besides being relatively easy to implement, it was already understood by the team-members (featured in Controls and Systems module). LQG requires a linearised model which was already being developed and even though robustness is not guaranteed, as defined in the objectives, stability was the priority.

3 Theoretical Background

LQG control is, essentially, the combination of a Linear Quadratic Regulator and a Kalman filter. Given a controllable state-space model A ($n \times n$) and B ($n \times m$), where n is the number of states and m the number of control inputs, by means of the feedback law in equation 1, the closed-loop system ($A - BK$) eigenvalues can be made arbitrarily stable. The choice of gain K and more detailed information on LQR and controllability can be found in [2].

$$u = -Kx \quad (1)$$

where u is the input vector and x the state vector. Even though LQR requires full-state measurements of the system, this is not always possible. The combination of an LQR and a full-state observer solves this issue, hence the importance of the Kalman filter.

3.1 Observability

A system is observable if its observability matrix shown below has full rank, meaning the rank is equal to the value n . The physical meaning of observability is related to the ability of full-estimating the states based on the available measurements.

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \quad (2)$$

3.2 Kalman filter

Provided that the state-space model pair A and C ($p \times n$), where p is the number of state measurements, is observable, the Kalman filter can estimate the full-state vector. The filter takes as inputs the previous estimate and the current input to predict the current state. On top of the prediction, the measurements are also taken into account. The following equations correspond to the Kalman filter mathematical algorithm which can be found on chapter 8 of Brunton & Kutz [3].

Starting from the state space system equations, w_d and w_n are added to the system as sensor noise and state disturbances.

$$\frac{d}{dt}x = Ax + Bu + w_d \quad (3)$$

$$y = Cx + Du + w_n \quad (4)$$

The equations below shows the relation between the expected value of noise and disturbances and their covariance matrices V_n and V_d . These matrices follow a Gaussian distribution and need to be tuned for each system.

$$\mathbb{E}(w_d(t) w_d(\tau^*)) = V_d \delta(t - \tau) \quad (5)$$

$$\mathbb{E}(w_n(t) w_n(\tau^*)) = V_n \delta(t - \tau) \quad (6)$$

Similarly to LQR, the Kalman filter gain can be calculated the solution Y to the Riccati equation:

$$YA^* + AY - YC^*V_n^{-1}CY + V_d = 0 \quad (7)$$

$$K_f = YC^*V_n \quad (8)$$

This method minimizes a cost function (9) that takes into account both noise and disturbance.

$$J = \lim_{t \rightarrow \infty} \mathbb{E}((x(t) - \hat{x}(t))^*(x(t) - \hat{x}(t))) \quad (9)$$

The gain K_f is then used to estimate the full-state vector $\hat{x}$.

$$\frac{d}{dt}\hat{x} = A\hat{x} + Bu + K_f(y - \hat{y}) \quad (10)$$

$$\hat{y} = C\hat{x} + Du \quad (11)$$

$$\frac{d}{dt}\hat{x} = (A - K_fC)\hat{x} + [K_f, (B - K_fD)] \begin{bmatrix} y \\ u \end{bmatrix} \quad (12)$$

Finally, it becomes clear below that, by considering the estimation error $\epsilon = x - \hat{x}$, the error converges to zero by making the eigenvalues of $(A - K_fC)$ stable:

$$\frac{d}{dt}\epsilon = \frac{d}{dt}x - \frac{d}{dt}\hat{x} = (A - K_fC)\epsilon + w_d - K_fw_n \quad (13)$$

3.3 LQG

The next equations incorporate the Kalman filter into LQR from [3]. K_r is the regulator gain, which, once again follows the Riccati equations by minimising the cost function $J(t)$.

$$\frac{d}{dt}\hat{x} = (A - K_fC - BK_r)\hat{x} + K_fy \quad (14)$$

$$u = -K_r\hat{x} \quad (15)$$

$$\frac{d}{dt}x = Ax - BK_r\hat{x} + BK_r\epsilon + w_d \quad (16)$$

$$\frac{d}{dt} \begin{bmatrix} x \\ \epsilon \end{bmatrix} = \begin{bmatrix} A - BK_r & BK_r \\ 0 & A - K_f C \end{bmatrix} \begin{bmatrix} x \\ \epsilon \end{bmatrix} + \begin{bmatrix} I & 0 \\ I & -K_f \end{bmatrix} \begin{bmatrix} w_d \\ w_n \end{bmatrix} \quad (17)$$

$$J(t) = \left\langle \int_0^t [x(\tau)^* Q x(\tau) + u(\tau)^* R u(\tau)] d\tau \right\rangle \quad (18)$$

What makes the implementation of LQG is so simple is the fact that the gain values of the controller and Kalman filter (K_r and K_f respectively) can be computed independently and yet optimizing the problem overall. Finally, the eigenvalues of the whole system correspond to both $(A - BK_r)$ and $(A - K_f C)$. Figure 1 shows a schematic representation of the implementation of an LQG controller (Figure 8.11 [3]).

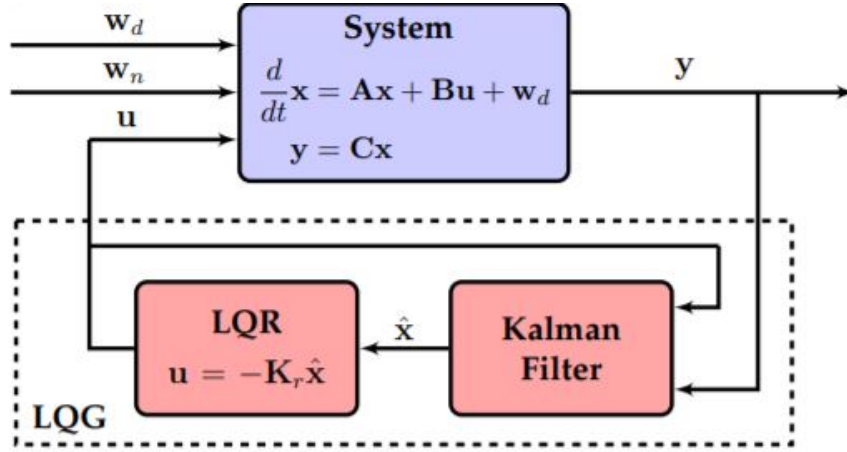


Figure 1: LQG controller schematic implementation.

4 Design and Implementation of LQG

The state-space model matrices A and B were computed for the most critical mission legs: hover with payload and cruise with payload. The considered states are displayed below and the equations of motion and linearization process are described in [6]. For practicality, the states are going to be referred to according to the vector order.

$$x = \begin{bmatrix} u & v & w & p & q & r & \phi & \theta & \psi & \phi_L & \theta_L & \dot{\phi}_L & \dot{\theta}_L \end{bmatrix}^T \quad (19)$$

4.1 Trajectory generation

Considering the autonomous part of the mission, one of the most important aspects of the controller was the ability of tracking trajectories. Initially, the plan was to integrate a trajectory generator into the closed-loop system as the reference value from which the error ϵ is computed. Figure 2 shows an example of the implementation.

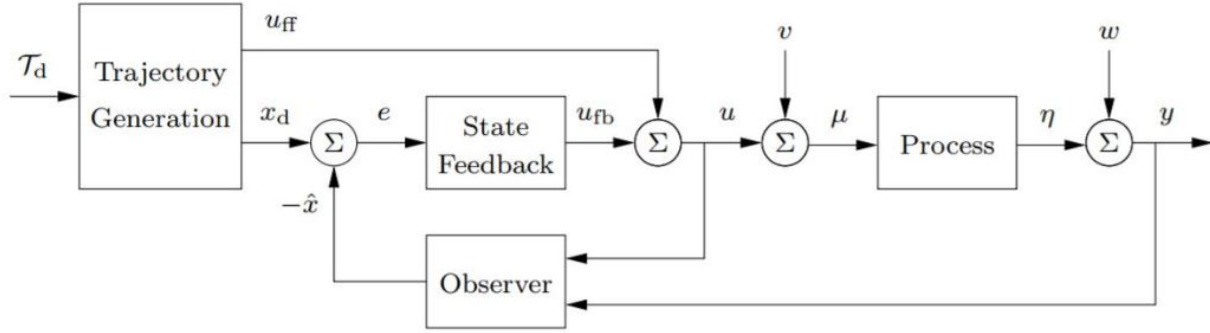


Figure 2: Incorporation of trajectory generation into the LQG controller.

In order to generate an optimal trajectory, mathematical methods were investigated, particularly Calculus of Variations. At the end the Control group decided that a trajectory generator would not be feasible within the given time-frame. The helicopter to be controlled has 6Dof, which make the trajectory planning extra complex. As a result, the focus was shifted from complicated trajectories, to simple step inputs. More specific details on the tests to assess the performance of the controller are described in [2].

4.2 Sensors

The decision on the sensors and navigation systems to be integrated into the helicopter was made by Systems and Propulsion team. Initially the team opted for a GPS and a set of inertial navigation system sensors made up of 3 accelerometers and 3 gyroscopes (Ellipse 2 series), which measure non-gravitational acceleration and angular rate, respectively. The implementation of Kalman filter on a helicopter was explored in [4] and considering the previous choices it was decided to follow a similar approach. The design of the filter is illustrated below (Figure 2.4-[4]). The measurements captured by the sensors are integrated through navigation algorithms to compute the attitudes, positions and velocities of the helicopter. Additionally, the GNSS (Global Navigation Satellite System) also provides information on position and velocity to be added to the filter. Having two sources of measurements is vital since sensor-based information may easily diverge and satellite-based information can be interrupted. The idea is that both measurements can calibrate each other.

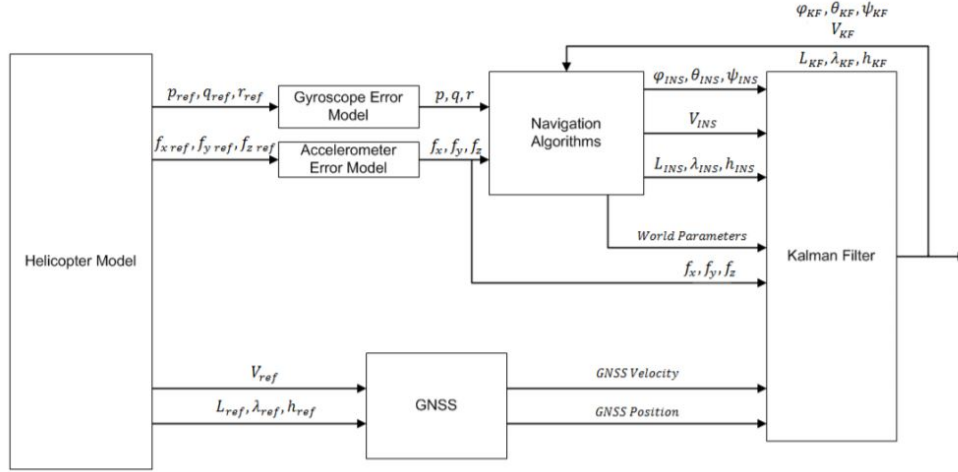


Figure 3: Kalman filter design with integrated sensor and GNSS measurements [4].

However, in collaboration with Systems and Propulsion team the GPS+INS choice was replaced with an integrated INS/GPS (Kearfott model) system. This decision simplified the filter design since it removed the need for a merging algorithm. The measurements from the new system are already calibrated and provide with the first 9 states (velocities, angular velocities and angles).

The last 4 states are related to the angular position and angular rate of the payload helicopter slung. The measurements of the angles (10th and 11th states) are provided by a lidar camera (Leddar Vu8) which estimates the distance between camera and payload.

4.3 Matlab implementation

Two separate Matlab codes were written for each mission leg to account for the LQR and Kalman filter separately so that the gains can be combined to create a Simulink simulation. From the state-space model, an augmented system was created in order to account for disturbance and noise. The kalman filter gain K_f was easily calculated by modifying slightly the '*lqr*' command on Matlab and incorporating the augmented system. As seen before, this is only possible considering that both the filter and LQR minimize an equivalent cost function. The command '*randn*' generates gaussian random numbers and it was used to simulate disturbances and noise on the system. The augmented systems needed the noise and disturbance covariance matrices V_n and V_d which are diagonal matrices of dimensions (11×11) and (13×13) , respectively. There are complex mathematical methods which attempt to accurately estimate the variance of noise and disturbance. In this case the values were based on the uncertainty of the sensors and cameras. It was assumed that the noise would interfere more than the disturbance and therefore the disturbance covariance resulted from an applied scaling factor 10^{-2} to the identity matrix whereas the diagonal values of the noise covariance are shown below.

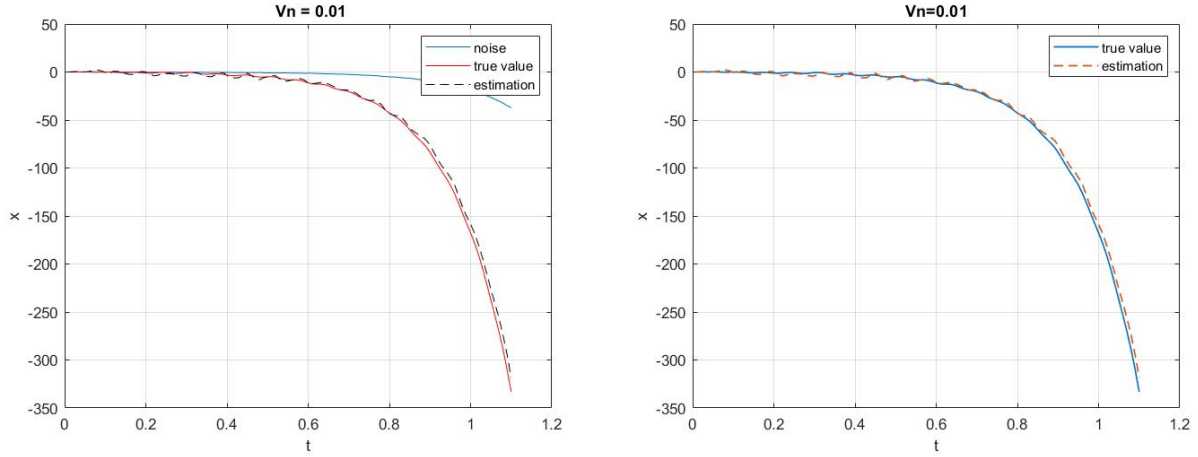
$$\begin{bmatrix} 10 & 10 & 10 & 100 & 100 & 100 & 10 & 10 & 10 & 10 & 10 \end{bmatrix} \quad (20)$$

5 Results

The observability matrix for the (A,C) pair for both mission legs was tested and the rank was found to be 9 in both cases, which does not correspond to full-state. This suggests that the system is not observable. However, the observability of each state was computed and it is displayed below.

From this data, it became clear that the most observable states are the last 5. The rank of the observability of the full system was computed again but this time assuming the last 5 states as the only measured states. This system was found to be observable. Since observability is a necessary condition to implement a Kalman filter, the rest of the results were found assuming a (5×13) C matrix just to test the filter design.

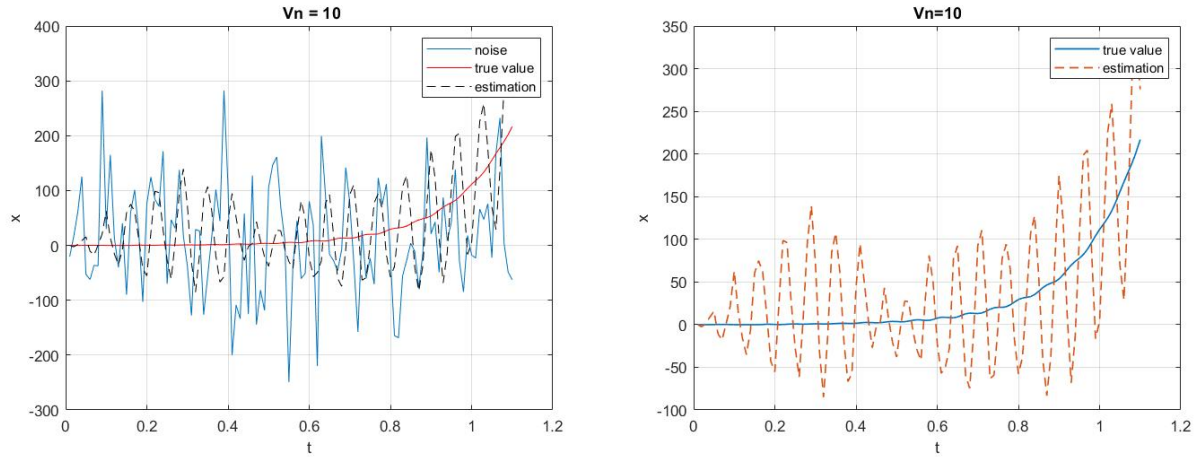
The set of plots below show the performance of the Kalman filter in response to an impulse input test, by comparing noise and estimated value to the true one. In all the cases below, V_n was kept as the identity matrix, except for the value correspondent to the state being measured. For example in figure 4, the 11th state's response was analysed and the noise was filtered by applying a 0.01 factor to the 11th state.



(a) Comparison between noisy measurement, true value and Kalman filter estimation of the state. (b) Comparison between true value and Kalman filter estimation of the state.

Figure 4: Result of tuning process of the 11th state with a covariance value of 0.01.

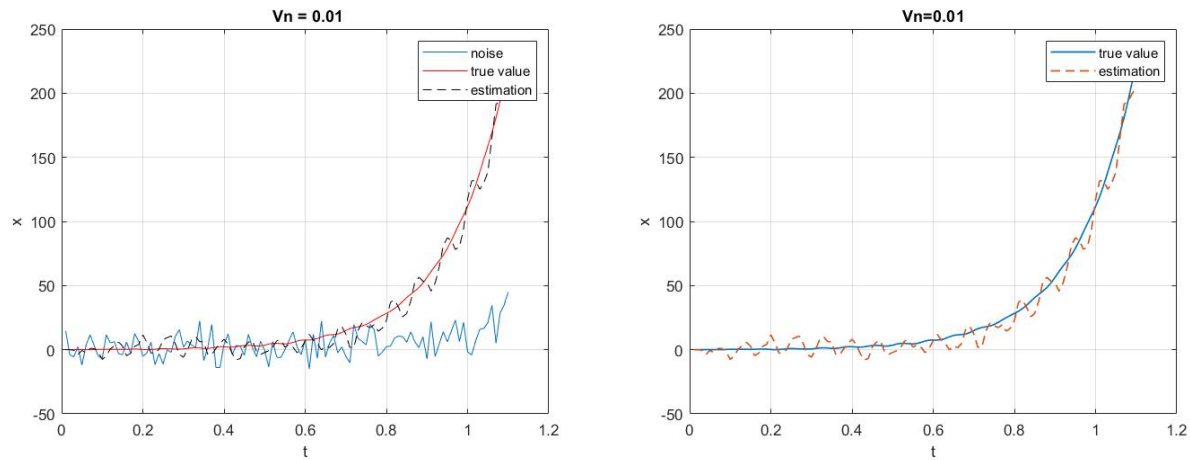
Similarly, the same process was repeated for the 13th state. Figure 5 and 6 show the comparison between a covariance value 10 and 0.01. Clearly in the second case, the system's ability to filter the noise was improved in the second case.



(a) Comparison between noisy measurement, true value and Kalman filter estimation of the state. (b) Comparison between true value and Kalman filter estimation of the state.

Figure 5: Result of tuning process of the 13th state with a covariance value 10.

Finally, in figure 7, the estimated values were plotted altogether as the response to the impulse signal. The estimated values were plotted as dashed lines but it is difficult to tell them apart.



(a) Comparison between noisy measurement, true value and Kalman filter estimation of the state. (b) Comparison between true value and Kalman filter estimation of the state.

Figure 6: Result of tuning process of the 11th state with a covariance value 0.01.

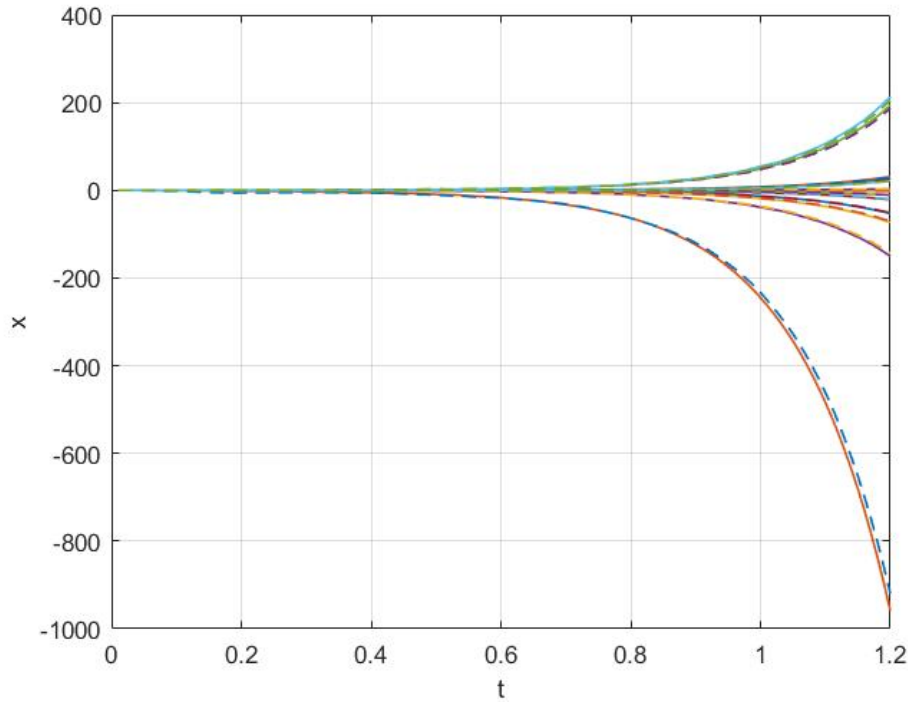


Figure 7: Kalman filter estimation for the full state.

6 Discussion

Defining the state-space model of the helicopter, which was developed by other members of the team, was a long and complex process. Since the LQG controller is dependent on the model, there was not enough time to implement and test the whole controller. It became clear that observability cannot be analysed as whole. The level of observability of a state needs to be considered before deciding on the sensors and systems to be incorporated. In this case the measurement of 5 specific states proved to be more powerful than the original 11 measurements planned. Analysing this aspect before could have potentially reduced costs. The Kalman filter appears to be an accurate way of estimating full-state and smoothing out noise and disturbance. However it is important to mention that these results do not represent the the system, since on the real case, the system was not observable.

7 Improvements

Since the project was time-constrained, perhaps a better strategy would have been assigning 2 people for control from the beginning of the project and choosing a control methodology that is not dependent on the the model (eg:MPC). If the controller had actually been designed, it would be interesting to investigate its robustness properties and explore Loop-transfer recovery.

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